

SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

CHENNAI – 602105

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Correlation and Pattern Analysis

Course Code:	DSA0502	Course Title:	Query Processing for Data Science
CO Assessed:	CO4	Assessment:	Tool 3– Correlation and Pattern Analysis
Weightage:	25%	Total Marks:	
CO4 Statement: Explore datasets using analytical techniques such as grouping, correlation detection, joining datasets, and extracting meaningful insights.			
Student Name:	BHARANI DHARAN S	Reg. No.:	192424141
Section Batch: /	D/2024	Date of Submission:	17/06/26
Faculty In-charge:	Dr. B.Shyamala Gowri	Project Mode:	Individual Submission

Code of Conduct

I BHARANI DHARAN S{192424141} (Name / Reg No) certify that this submission is my original work and that I have adhered to all guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature: BHARANI DHARAN S

SALES AND CUSTOMER PATTERN ANALYSIS

AIM

To perform **Sales and Customer Pattern Analysis** by generating and integrating customer and sales transaction datasets, exploring the data, analyzing correlations and sales trends, identifying customer segments using **K-Means clustering**, detecting outliers using **Isolation Forest**, and presenting the results through suitable visualizations.

ALGORITHM

1. **Start the program.**
2. **Import the required libraries** such as Pandas, NumPy, Matplotlib, Seaborn, StandardScaler, K-Means, and Isolation Forest.
3. **Generate the customer dataset** containing:
 - Customer ID
 - Age
 - Gender
 - Income level
 - Customer tenure
 - Purchase frequency
 - Discount preference
4. **Generate the sales transaction dataset** containing:
 - Transaction ID
 - Customer ID
 - Purchase date
 - Purchase amount
 - Discount applied
 - Product category
 - Satisfaction score
 - Payment method
5. **Merge the customer and sales datasets** using customer_id.
6. **Perform Exploratory Data Analysis (EDA)** by calculating descriptive statistics and checking for missing values.
7. **Perform correlation analysis** between age, purchase frequency, discount, purchase amount, discount preference, and satisfaction score.
8. **Analyze sales patterns** based on product category, payment method, customer tenure, and month.
9. **Create visualizations** such as:
 - Correlation heatmap
 - Age vs. purchase amount
 - Income level vs. purchase amount
 - Purchase frequency distribution
 - Category-wise sales
 - Age distribution
 - Discount vs. satisfaction
 - Payment method analysis
 - Monthly sales trend
10. **Prepare customer-level data** by aggregating transaction information for each customer.
11. **Standardize the selected customer features** using StandardScaler.
12. **Apply the Elbow Method** to determine a suitable number of clusters.
13. **Apply K-Means clustering** with four customer segments.

- [illegible]

